

What Is a Computer?

Lesson 01 — Fundamentals & Data Representation

JKSSB Computer Awareness

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Video Lesson 01

Watch: What is a Computer?

Start with a Familiar Task

- ▶ A clerk enters marks into a spreadsheet.
- ▶ A formula totals the marks and calculates a percentage.
- ▶ The screen shows the result; the file can be saved for later.

Question

Which parts of this task are input, processing, output, and storage?

A Working Definition

Computer

An electronic, programmable system that accepts data, processes it under instructions, produces information, and can store data or results.

- ▶ **Data:** raw facts or symbols supplied to a system.
- ▶ **Information:** data organized or processed to be useful.
- ▶ A computer follows instructions; it does not guarantee that the supplied data or chosen instructions are correct.

Data Becomes Information

Data

- ▶ 68, 74, 81
- ▶ Three separate marks
- ▶ Not yet a summary

Information

- ▶ Total = 223
- ▶ Average = 74.33
- ▶ Results now answer a question

The same values can serve as data in one task and information in another, depending on the question being asked.

The IPO Cycle

1. **Input**: data and instructions enter through input devices or software.
2. **Processing**: the CPU and programs transform the data.
3. **Output**: results are presented through a display, printer, speaker, or another output device.
4. **Storage**: data and results are kept for current or later use.

Exam cue

Storage is often shown alongside the Input–Process–Output (IPO) model; the four-part view is commonly called IPOS.

Characteristics: What Computers Do Well

- ▶ **Speed**: perform many operations quickly.
- ▶ **Accuracy**: repeat calculations consistently when data and instructions are correct.
- ▶ **Diligence**: do not tire from repetitive work.
- ▶ **Storage**: retain large amounts of data in a compact form.
- ▶ **Versatility**: support different tasks by running different programs.
- ▶ **Automation**: continue a programmed task once started and supplied with required input.

Limitations: What Computers Do Not Decide

- ▶ A computer does not independently determine whether an input is truthful or fair.
- ▶ Incorrect data or instructions can produce incorrect results.
- ▶ It has no human judgement or common-sense understanding by default.
- ▶ Output still needs interpretation by a person in context.

Remember

GIGO means *Garbage In, Garbage Out*: poor input can lead to poor output.

Applications Around Us

- ▶ **Office**: documents, calculations, records, and communication.
- ▶ **Banking**: account records, transaction processing, and ATMs.
- ▶ **Education**: learning platforms, assessments, and digital resources.
- ▶ **Healthcare**: patient records, appointments, and diagnostic support.
- ▶ **Governance**: online forms, records, and citizen services.

The application changes; the input–processing–output idea remains.

Quick Classification Drill

- ▶ Raw marks entered into a form: **data / input**.
- ▶ A formula calculating an average: **processing**.
- ▶ The average shown on screen: **output / information**.
- ▶ A saved spreadsheet: **storage**.

Takeaway

Identify the role in the current task. Do not classify a file as processing, or a screen as storage, merely because both are used by a computer.

Lesson 01: Recall

- ▶ A computer accepts data, follows instructions, processes, outputs, and can store.
- ▶ Data are raw facts; information is useful processed or organized data.
- ▶ IPO = Input, Processing, Output; storage is the additional S in IPOS.
- ▶ Speed and accuracy are strengths; correctness depends on input and instructions.